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Jane Smith

Chapter 1
Introduction

Underwater inspection and maintenance often require physical interaction with submerged structures. These tasks are commonly performed by divers or by remotely operated vehicles, which is expensive and slow. This chapter introduces the problem, the goals of the thesis and the methods that were used to reach them in the laboratory.

References

[1] G. Marani, S. K. Choi and J. Yuh, Underwater autonomous manipulation for intervention missions, Ocean Engineering, 2009.

[2] M. Prats, D. Ribas and N. Palomeras, Reconfigurable AUV for intervention missions, Intelligent Service Robotics, 2012.

Appendix A
Mechanical Component Drawings

This appendix presents the engineering drawings of the payload and the passive hook that were used in the experiments described in the previous chapters.

Thruster model

Each thruster produces a force along its axis. The forces of all thrusters are collected in a vector and mapped to the body frame.

$$\tau = Bf, \tag{2.13}$$

The matrix B depends on the position and direction of each thruster. The requested wrench may not always be achievable. Each thruster has a limited output and saturation must be considered.

Table 4.5: Implemented work and its purpose.

Goal	Implemented work	Purpose and boundary
Passive line placement	Printed hook, passive link, mounting, and matching payload handle.	Provide the interaction mechanism without a powered arm.
Planned approach	Recorded target pose, A*, path processing, and time-dependent references.	Generate a map-checked route to the pre-hook pose. Final interaction retained a fixed direction.

$$\begin{aligned} \min \quad & J(u) \\ \text{subject to} \quad & x = f(x,\, u), \\ & u_{\min} \le u \le u_{\max} \end{aligned} \tag{2.35}$$

Here, N is the prediction horizon length and the vector x is the latest state estimate.

This agrees with Smith *et al.* measured *in situ* with weights **w**.

Table 5.3: Endpoint estimates.

Trial	Last position error (cm)	Near-limit samples (%)
Near-front	3.39	0.0
Right-side	4.79	49.8
Direct-front	8.00	64.8